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If the application for this patent was filed on or after December 12, 1980, maintenance fees are due three years and six months, seven years and six months, and eleven years and six months after the date of this grant, or within a grace period of six months thereafter upon payment of a surcharge as provided by law. The amount, number and timing of the maintenance fees required may be changed by law or regulation. Unless payment of the applicable maintenance fee is received in the United States Patent and Trademark Office on or before the date the fee is due or within a grace period of six months thereafter, the patent will expire as of the end of such grace period.

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If the application for this patent was filed on or after June 8, 1995, the term of this patent begins on the date on which this patent issues and ends twenty years from the filing date of the application or, if the application contains a specific reference to an earlier filed application or applications under 35 U.S.C. 120, 121, 365(c), or 386(c), twenty years from the filing date of the earliest such application ("the twenty-year term"), subject to the payment of maintenance fees as provided by 35 U.S.C. 41(b), and any extension as provided by 35 U.S.C. 154(b) or 156 or any disclaimer under 35 U.S.C. 253.

If this application was filed prior to June 8, 1995, the term of this patent begins on the date on which this patent issues and ends on the later of seventeen years from the date of the grant of this patent or the twenty-year term set forth above for patents resulting from applications filed on or after June 8, 1995, subject to the payment of maintenance fees as provided by 35 U.S.C. 41(b) and any extension as provided by 35 U.S.C. 156 or any disclaimer under 35 U.S.C. 253.



(12) **United States Patent**
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(54) **BILLING SYSTEM BASED ON RADON CONCENTRATION PATTERN ANALYSIS AND METHOD THEREFOR**

(71) Applicant: **FTLAB co., LTD.**, Ansan-si (KR)

(72) Inventors: **Jae-Jun Ko**, Seoul (KR); **Young-Gweon Kim**, Anyang-si (KR)

(73) Assignee: **FTLAB CO., LTD.**, Ansan-si (KR)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 129 days.

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Primary Examiner — Rokib Masud
(74) *Attorney, Agent, or Firm* — ArentFox Schiff LLP

(57) **ABSTRACT**

A billing system based on radon concentration pattern analysis and a method therefor is provided, the system including a radon detection device counting the output alpha particle detection signals for a predetermined measurement time to calculate a radon concentration value; and a cloud server storing generating a radon concentration pattern corresponding to the radon detection device by using a radon concentration value received from the radon detection device and calculating a cost for radon monitoring service provided to a terminal of the user, wherein the cloud server analyzes the generated radon concentration pattern and determines the periodic behavior pattern of the user for a predetermined period of time, whereby it is possible to charge a radon monitoring service fee for each user using the radon monitoring service, and the user may be motivated to frequently perform the ventilation action that reduces the radon concentration.

15 Claims, 10 Drawing Sheets

